

Data Cleaning & Preprocessing – Summary

Loaded the e-commerce shopping dataset (1000 rows, 19 columns) into a Pandas DataFrame and explored it using `head()`, `tail()`, `info()`, and `describe()` to understand structure, datatypes, and missing values.

Calculated missing-value percentages per column and dropped five columns `what_customers_said`, `videos`, `variations`, `seller_information`, and `seller_name` as each had more than 30% missing data and couldn't be reliably imputed. The `discount` column (~12% missing) was filled using the median.

Cleaned the `final_price` column by stripping the ₹ symbol and commas and converting it to numeric; rows where the value still couldn't be parsed were filled using `initial_price` as a fallback.

Cleaned up the `sizes` field (extracted readable size labels from nested data) and standardized `best_offer` so empty values show as "No best offer available."

Selected the relevant columns for analysis (`id`, `title`, `category`, `pricing`, `discount`, `sizes`, `offers`, `rating`, `ratings count`) and filtered out rows with missing or invalid `final_price`.

Checked for and removed duplicate records.

Exported the final cleaned dataset as `cleaned_dataset.csv`.